

Computer Vision Practical Report:

8-NEIGHBOUR LAPLACIAN MASK

1. AIM

To sharpen an image by applying an extended 8-neighbor Laplacian mask with diagonal elements in OpenCV, display the original and sharpened images, and save the output.

2. PROCEDURE

- Load the image and convert it to grayscale.
- Define an 8-neighbor Laplacian kernel including diagonal elements (e.g., center = -8 , all 8 neighbors = $+1$).
- Filter the image with the 8-neighbor kernel using `cv2.filter2D()`.
- Subtract the Laplacian response from the original image to obtain the sharpened output.
- Display the original and sharpened images side-by-side using Matplotlib, and save the result using `cv2.imwrite()`.

3. PROGRAM CODE

```
import cv2
import numpy as np

image = cv2.imread(r"C:\Users\adminuser\Desktop\Python CV\E21-IMG.png")

gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# Laplacian mask with diagonal neighbors
kernel = np.array([
    [1, 1, 1],
    [1, -8, 1],
    [1, 1, 1]
], dtype=np.float32)
```

```
# Apply Laplacian mask
laplacian = cv2.filter2D(gray, cv2.CV_32F, kernel)

# Sharpen image
sharpened = gray - laplacian

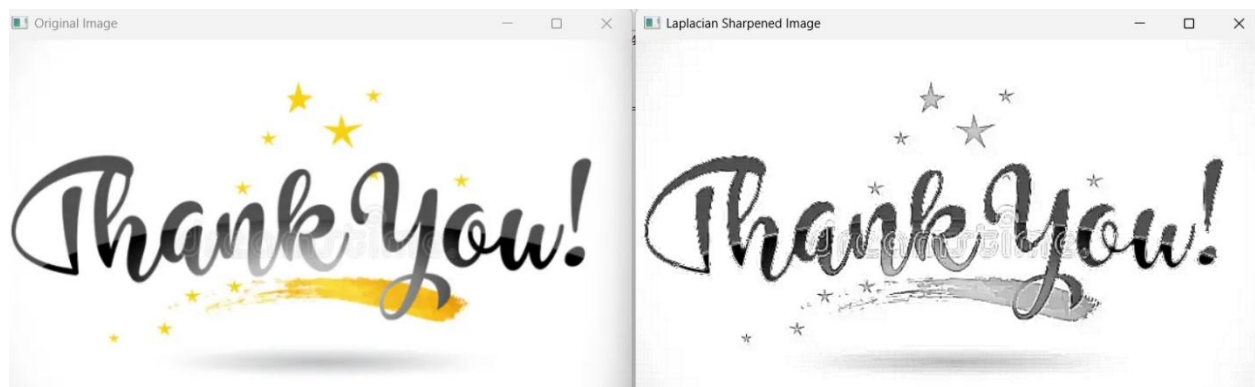
# Convert to 8-bit
sharpened = cv2.convertScaleAbs(sharpened)

cv2.imshow("Original Image", image)
cv2.imshow("Laplacian Sharpened Image", sharpened)

cv2.imwrite("laplacian_diagonal_sharpened.jpg", sharpened)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

4. OUTPUT



5. RESULT

The Python script executed successfully. The input image was loaded.

